

Music Visualizer

Introduction:

Adds a visual aspect to listening to music through LED lights. It corresponds the mood of the song to the lights in live time. This allows for an enhanced experience by incorporating live responses to audio. By utilizing aspects of songs like volume, notes, and time intervals between notes, updates to the LEDs and their colors are made arranging an appealing visual array of lights.

Results:

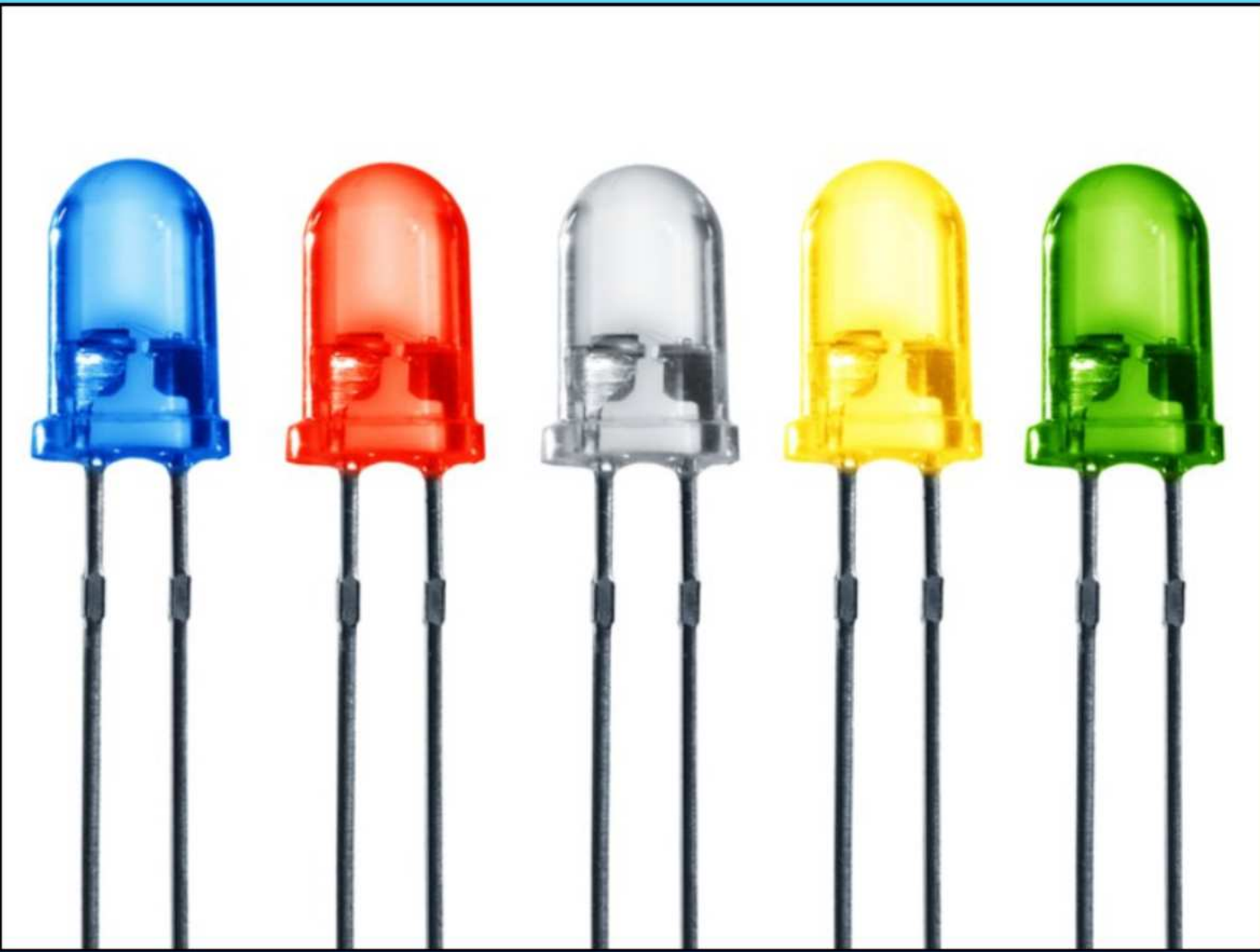
Successfully developed an Arduino circuit with live light updates and audio input. Using tempo, notes, and volume, the lights are updated and the colors shift. Tests were conducted to test the accuracy of the note recognition using a piano as seen in the table below. However, no quantative testing was done on the accuracy of matching the moods of the songs.

Conclusion:

Using live note recognition is a viable method to enhance the music listening experience. With upscaling and more development, the music visualizer can be an excellent way to make music more enjoyable. Further development in testing would be needed to create a more accurate model and method for determining the mood of songs in their current moment especially if implemented with improvised music performances.

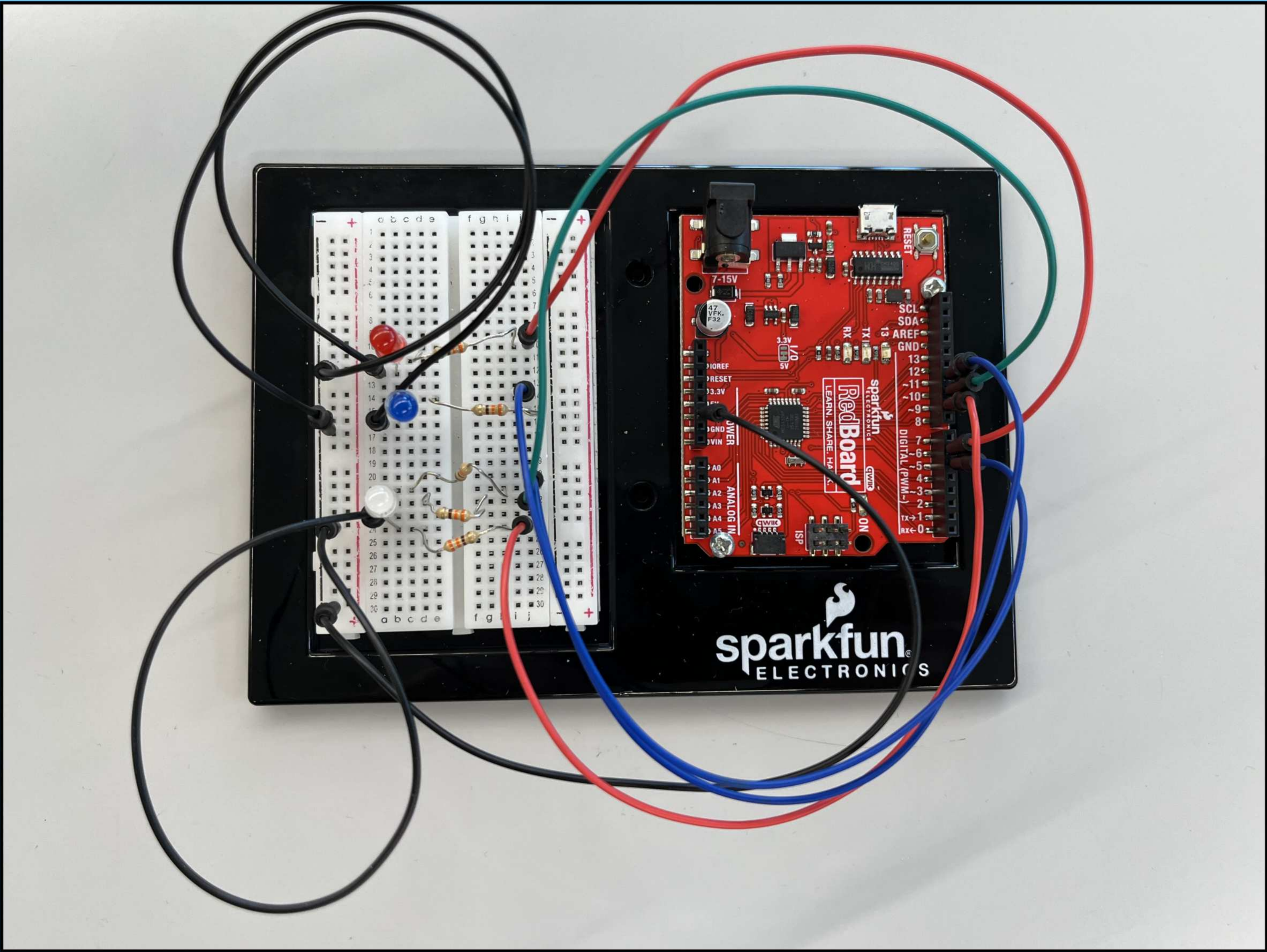
Octave	Notes											
Number	C	C#	D	D#	E	F	F#	G	G#	A	A#	B
0	0	1	2	3	4	5	6	7	8	9	10	11
1	12	13	14	15	16	17	18	19	20	21	22	23
2	24	25	26	27	28	29	30	31	32	33	34	35
3	36	37	38	39	40	41	42	43	44	45	46	47
4	48	49	50	51	52	53	54	55	56	57	58	59
5	60	61	62	63	64	65	66	67	68	69	70	71
6	72	73	74	75	76	77	78	79	80	81	82	83
7	84	85	86	87	88	89	90	91	92	93	94	95
8	96	97	98	99	100	101	102	103	104	105	106	107
9	108	109	110	111	112	113	114	115	116	117	118	119
10	120	121	122	123	124	125	126	127	-	-	-	-

A	B	C	D	E	F	G
10/10	10/10	9/10	10/10	9/10	10/10	10/10



Methods:

Through Arduino and Python, a computer can take in audio and read the music that is being played in live time. The program then uses music theory to determine the mood and change the lights. Music theory concepts like semitones shifts (note jumps), tempo (the general pace of the music), volume (measured in decibels) to change the colors displayed



References:

H. Babbar and B. Singhal, "Real time staff notation generation of guitar along with standard guitar tuner," 2011 International Conference on Energy, Automation and Signal, Bhubaneswar, India, 2011, pp. 1-5, doi: 10.1109/ICEAS.2011.6147189. keywords: {Tuners;Mathematical model;Equations;Real time systems;Computers;real-time staff notation generation;standard tuner based on MIDI number;staff notation;musical pitches on a guitar},

Miller, A. (2021, November 24). What makes A happy song happy?. Piano Lessons, Voice, Guitar, Violin, Drums. <https://www.creativesoulmusic.com/blog/what-makes-a-happy-song-happy>